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DECENTRALIZED RENEWABLE POWER GENERATION AND METHODS FOR UNIVERSAL BASIC INCOME

Abstract

A device for generating power may include a mobius magnet array. A device may include an external magnet array configured to generate a magnetic field within the device, wherein the magnetic field exerts a rotational force on the mobius magnet array. A device may include a shaft connected to the mobius magnet array, the shaft configured to rotate when the magnetic field exerts a force on the mobius magnet array. A device may include an alternator connected to the shaft, the alternator configured to convert the rotational motion of the shaft into electrical power.

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Background/Summary

BACKGROUND

[0001] Most viable commercial electric power generation systems today share common and limiting characteristics. Coal, liquid combustible, and nuclear power generation systems require heat or chemical reactions and produce pollutants as a byproduct. These systems are not necessarily renewable or clean. The electrical output of these systems is heavily centralized and the distribution infrastructure (transmission poles) is unreliable, susceptible to failure, and expensive to maintain. So called alternative clean energy systems such as solar and wind require limited elements and are not viable in all environmental locations. They also struggle to meet on-demand energy needs without supplemental battery storage systems.

[0002] With the exception of solar energy, today's commercial power generation systems convert mechanical energy into electrical output. A nuclear power plant for example is an expensive and dangerous steam engine that creates rotational inertia and converts that mechanical energy into electricity. A gas powered generator utilizes a combustion motor to mechanically rotate a shaft to produce an electrical output. In its simplest form mechanical rotation about a point can be converted into electrical energy. The rotating inertia mechanism or mechanical energy produced by these systems are commonly called prime movers. Prime movers convert the primary source into mechanical energy that create electrical energy when combined with some kind of alternator or magnetic/electric induction machine. These existing systems are utilizing limited energy sources to power their prime movers to generate mechanical rotation. Re-thinking the source of mechanical rotation is the novelty of the described design in this patent.

SUMMARY

[0003] In some aspects, the techniques described herein relate to a device for generating power, the device including: a mobius magnet array including: a surface in the form of a mobius strip; and a plurality of magnets disposed on the surface; an external magnet array configured to generate a magnetic field within the device, wherein the magnetic field exerts a rotational force on the mobius magnet array; a shaft connected to the mobius magnet array, the shaft configured to rotate when the magnetic field exerts a force on the mobius magnet array;

[0004] and an alternator connected to the shaft, the alternator configured to convert the rotational motion of the shaft into electrical power.

[0005] In some aspects, the techniques described herein relate to a method for generating power, the method including: providing a mobius magnet array including: a surface in the form of a mobius strip; and a plurality of magnets disposed on the surface; exposing the mobius magnet array to an external magnetic field; rotating the mobius magnet array in response to the external magnetic field; rotating a shaft connected to the mobius magnet array; and converting the rotational motion of the shaft into electrical power.

[0006] In some embodiments, the method further includes: directing at least a portion of the electrical power to hardware; providing computational power, via the hardware, to a blockchain network; and validating a blockchain within the blockchain network.

[0007] In this outlined design the mechanical rotation needed to output electricity utilizes permanent magnets placed in a fixed array on a mobius strip or band (twisted cylinder) to force helical or rotational direction of a shaft. The positive and negative poles of permanent magnets when placed in a specific array on a mobius band surrounded by additional magnets can produce a unidirectional push-pull force and rotate a shaft. The unique profile of a mobius strip can align the push-pull forces of individual magnets to create the angular velocity of the shaft. This rotation is used to create multi-phase electricity when integrated with a traditional rotating electric generator. The output of this permanent magnet array can create enough electricity to power much larger

electro-magnetic arrays on similar mobius strips rotating larger shafts for increased output.

[0008] This architecture can be scalable to any size from small consumer electronic devices to warehouse sized power generators supplying clusters of neighborhoods. The rotating magnet mobius array can be buried underground as it would require little maintenance. This mechanical energy source is permanent, clean, unlimited, and no heat or burning is required. This motion would require little to no maintenance except for lubrication of bearings and shafts where rotation is occurring. Power generation can become decentralized and can be isolated to an electronic device, vehicle, building or home. No charging of battery or energy storage would be required. Energy consumption would be on demand. The need for large public infrastructure would become obsolete and power generation and distribution would become significantly more robust as the grid becomes decentralized. Power generation exists at the point location where the consumer needs it.

[0009] When power generation becomes decentralized and unlimited in supply the excess power available when the device, vehicle or home is not in use can be utilized by computer hardware like CPUs, graphical processors (GPUs), solid state storage devices (SSD), and radio antennas. Devices, vehicles, and homes can utilize the excess electrical energy to “mine” and “validate” blockchain based cryptocurrencies to earn monetary rewards. Similar to corporate server farms providing cloud computer services, a home or building becomes a competitive and reliable node in the network with resilient power backup redundancy.

[0010] Every cell phone, vehicle, and smart home device equipped with unlimited power supply and computational hardware can act as a dynamic validation node for large cryptocurrency networks to generate universal basic income. Blockchain based cryptocurrencies like Bitcoin and others would benefit from a vast and distributed network of computational power. Proof of work, proof of capacity, proof of stake cryptocurrencies reward participants with currency when contributing computational and electrical power to secure and validate their networks. Examples of blockchain mining include those in U.S. Pat. No. 11,196,566, which describes validating and mining of blockchain network with input from or participation of a third party, US Patent Pub. No. 2024/0005350 A1, and U.S. Pat. No. 10,771,524, which describe using hardware computational resource sharing in blockchains networks. Each of these references are hereby incorporated by reference in their entireties.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] The detailed description is set forth with reference to the accompanying figures. The use of the same numbers in different figures indicates similar or identical items.

[0012] For this discussion, the devices and systems illustrated in the figures are shown as having a multiplicity of components. Various implementations of devices and/or systems, as described herein, may include fewer components, and remain within the scope of the disclosure.

Alternatively, other implementations of devices and/or systems may include additional components, or various combinations of the described components, and remain within the scope of the disclosure.

[0013] FIG. 1 depicts a schematic view of a power generation system.

[0014] FIG. 2 depicts a schematic view of a steam turbine prime mover.

[0015] FIG. 3 depicts a schematic view of a wind powered turbine prime mover.

[0016] FIG. 4 depicts a schematic view of a cylinder and a mobius strip.

[0017] FIG. 5 shows a schematic view of a mobius magnet array.

[0018] FIG. 6A shows a schematic view of a device for generating power.

[0019] FIG. 6B shows a schematic view of a device for generating power connected to a battery storage system.

[0020] FIG. 7 shows a schematic view of a device for generating power.

[0021] FIG. 8A shows a schematic view of a power generation system.

[0022] FIG. 8B shows a schematic view of a system for generating power including a battery storage system.

[0023] FIG. 9A shows a schematic view of a system for powering a home.

[0024] FIG. 9B shows a schematic view of a system for powering a home with various power sources.

[0025] FIG. 9C shows a schematic view of a system for powering a home with various power sources.

[0026] FIG. 9D shows a schematic view of a system for powering a home with various power sources.

[0027] FIG. 10 shows a variety of forms that a power generation system can be implemented into
DETAILED DESCRIPTION

[0028] In the following detailed description, reference is made to the accompanying drawings, which form a part hereof. In the drawings, similar symbols typically identify similar components, unless context dictates otherwise. Thus, in some embodiments, part numbers may be used for similar components in multiple figures, or part numbers may vary from figure to figure. The illustrative embodiments described herein are not meant to be limiting. Other embodiments may be utilized, and other changes may be made, without departing from the spirit or scope of the subject matter presented here. It will be readily understood that the aspects of the present disclosure, as generally described herein, and illustrated in the Figures, can be arranged, substituted, combined, and designed in a wide variety of different configurations, all of which are explicitly contemplated and make part of this disclosure.

[0029] The following detailed description is directed to certain specific embodiments of the development. Reference in this specification to “one embodiment,” “an embodiment,” or “in some embodiments” means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the disclosure. The appearances of the phrases “one embodiment,” “an embodiment,” or “in some embodiments” in various places in the specification are not necessarily all referring to the same embodiment, nor are separate or alternative embodiments necessarily mutually exclusive of other embodiments. Moreover, various features are described which may be exhibited by some embodiments and not by others. Similarly, various requirements are described which may be requirements for some embodiments but may not be requirements for other embodiments. Furthermore, embodiments of the development may include several novel features, no single one of which is solely responsible for its desirable attributes or which is essential to practicing any particular embodiment described herein.

[0030] FIGS. 1-3 depict various devices commonly used for generating power by means of a prime mover. FIG. 1 depicts a schematic view of a power generation system 100. An energy source 110, such as nuclear power, a combustible fuel, wind, hydro, etc. Energy from the energy source 110 is provided to a prime mover 120 and the energy is converted into mechanical rotational movement and which in turn causes a shaft 126 of the prime mover to rotate. As seen in FIG. 1, the shaft 126 of the prime mover 120 is connected to an alternator 130 or generator. The alternator 130 is configured to convert the mechanical energy provided by the rotational movement of the shaft 126 into electrical power. The alternator 130 is then configured to transmit electrical power to a desired location. FIG. 2 depicts a schematic view of a power generation system 100 similar to FIG. 1. In FIG. 2, the energy source 110 is steam, typically high pressure steam. The steam can be produced by means of nuclear power or by means of burning coal. The steam is then forced into a steam turbine 112 which expands the steam and converts the energy received from the steam into mechanical energy which causes the shaft 126 to rotate. The shaft 126 is connected to a generator or alternator which converts the rotational motion of the shaft into electrical power. FIG. 3 shows a schematic view of a power generation system similar to FIGS. 1 and 2. In FIG. 3, the energy source

110 is wind. The wind causes rotation of blades **114** or other similar object that is connected to a shaft **126**. The rotation of the blades **114** and causes rotation of the shaft **126** similar to the methods described above. The shaft **126** is connected to a generator or alternator which converts the rotational motion of the shaft into electrical power.

[0031] FIG. **4** shows a cylindrical band **410** and a mobius strip **420**. A mobius strip or mobius as used herein refers to a continuous, one-side surface form by twisting one end of a rectangular strip through 180° about the longitudinal axis of the strip and attaching this end to the other. As seen in FIG. **4**, a cylinder **410** has a first surface **412** and a second surface **414** which form the cylinder **410**. The mobius strip **420** differs from the cylinder **410** in that the mobius strip **420** has a single continuous surface **422**.

[0032] FIG. **5** shows a schematic view of a mobius magnet array **530**. The mobius magnet array **530** includes a surface **532** and a plurality of magnets **534**. The surface **532** is formed as a mobius strip. The magnets **534** are disposed across the surface **532**. In some embodiments, the magnets **534** may be permanent magnets or electromagnets. In some embodiments, the magnets **534** are formed of a ferromagnetic material, such as, but not limited to, iron, nickel, cobalt, or combinations of the aforementioned materials. As seen in FIG. **5**, the magnets **534** may be disposed along the surface **532** of the mobius strip such that magnets **534** are oriented consistently relative to the surface profile of the mobius strip. Such an orientation rotates the direction of the magnetic poles of each magnet **534** as the surface **532** of the mobius strip also rotates. This orientation of magnets **534** beneficially propels helical or rotational motion of the mobius strip when exposed to an external magnetic field. In some embodiments, the magnets **534** are disposed on the surface **532** of the mobius strip in a spatially rotating pattern, such that the magnets **534** form an Halbach array. As used herein, a Halbach array refers to a special arrangement of magnets that augments the magnetic field on one side of the array while cancelling the field to near zero on the other side. A Halbach array may take the form of a variety of shapes including, but not limited to, a linear array of magnets, a variable linear array of magnets, or a cylinder.

[0033] Orienting the magnets in this orientation beneficially augments the magnetic field generated by the magnets **534** on a first side of the magnet array formed by the magnets **534**, while substantially canceling the magnetic field generated by the magnet array on a second side of the magnet array. This orientation of magnets **534** beneficially propels helical or rotational motion of the mobius strip when exposed to an external magnetic field.

[0034] In some embodiments, the surface **532** is formed of magnetic materials, such as ferromagnetic materials. In some embodiments, the surface **532** is formed of a Halbach array strip or Halbach array cylinder which is shaped into mobius band or strip. In some embodiments, the magnetic material of the surface **532** is magnetized. In some embodiments, the polarity of the magnetic surface is programmed to form a Halbach array.

[0035] FIG. **6** shows a schematic view of a device **600** for generating power. As seen in FIG. **6**, the system may include an outer shell **610** or surface, a plurality of external magnet arrays **620a** and **620b**, a mobius magnet array **630**, and a shaft **640**. As seen in FIG. **6**, the outer shell **610** may be cylindrical in shape. In some embodiments, the outer shell **610** encloses the shaft **640**, the external magnet array **620**, and the mobius magnet array **630**. In some embodiments, the outer shell **610** only encloses a portion of the shaft **640**.

[0036] The device **600** includes external magnet arrays **620a** and **620b**. In some embodiments, the magnets forming the external magnet arrays **620a** and **620b** may be permanent magnets or electromagnets. In some embodiments, the magnets forming the external magnet arrays **620a** and **620b** are formed of a ferromagnetic material, such as, but not limited to, iron, nickel, cobalt, or combinations of the aforementioned materials. The external magnet arrays **620a** and **620b** may be attached to the outer shell. In some embodiments, the external magnet arrays **620a** and **620b** are attached to an internal side of the outer shell **610**.

[0037] In some embodiments, the device **600** includes a first external magnet array **620a** and a

second external magnet array **620b**. In some embodiments, the first external magnet array **620a** is disposed at a first location on the surface of the outer shell **610**. In some embodiments, the second external magnet array **620b** is disposed at a second location on the surface of the outer shell **610**. In some embodiments, the first external magnet array **620a** is disposed on a first external surface of the outer shell **610** and the second external magnet array **620b** is disposed on a second external surface on the outer shell **610**. In some embodiments, the first and second magnet array **620a** and **620b** are orientated such that they are located opposite to each other. In some embodiments, the first and second external magnet arrays **620a** and **620b** are located on opposite sides of the outer shell **610** such that the mobius magnet array **630** and the shaft **640** are disposed between the first external magnet array **620a** and the second external magnet array **620b**. In some embodiments, the magnets of the first external magnet array **620a** are orientated on a surface of the outer shell **610** such that the magnetic poles of each magnet in the first external magnet array **620a** are arranged in the same direction. In some embodiments, the magnets of the first external magnet array **620a** are orientated such that the magnetic poles of at least one magnet in the array **620a** is aligned differently from an adjacent magnet in the array **620a**. In some embodiments, the first external magnet array **620a** creates a magnetic field within the interior of the outer shell **610**.

[0038] In some embodiments, the magnets of the second magnet array **620b** are orientated on a surface of the outer shell **610** such that the magnetic poles of each magnet forming the second external magnet array **620b** are arranged in the same direction. In some embodiments, the magnets of the second external magnet array **620b** are orientated such that the magnetic poles of at least one magnet in the array **620b** is aligned differently from an adjacent magnet in the array **620b**. In some embodiments, the magnets of the second external magnet array **620b** are orientated such that the poles of the magnets forming the second external magnet array **620b** are arranged opposite to the poles of the magnets forming the first external magnet array **620a**. In some embodiments, the second external magnet array **620b** creates a magnetic field within the interior of the outer shell **610**. In some embodiments, the magnetic field created by the second external magnet array **620b** overlaps with the magnetic field created by the first external magnet array **620a**. In some embodiments, the magnetic field created by the second external magnet array **620b** does not overlap with the magnetic field created by the first external magnet array **620a**.

[0039] The device **600** includes a mobius magnet array **630**. In some embodiments, the mobius magnet array **630** is similar or identical to the mobius magnet array **530** as described above.

[0040] The mobius magnet array **630** is disposed within the outer shell **610**. In some embodiments, the mobius magnet array **630** is disposed within the outer shell **610** such that the mobius magnet array **630** is located between the first external magnet array **620a** and the second magnet array **620b**. In some embodiments, the mobius magnet array **630** is similar or identical to the mobius magnet array **530** as described above. In some embodiments, the mobius magnet array **630** is orientated within the outer shell **610** such that the magnets of the mobius magnet array **630** are affected by the magnetic fields generated by the first and second external magnet arrays **620a** and **620b**. In some embodiments, the magnetic fields generated by the first and second external magnet exert a force on the magnets of the mobius magnet array **630**, thereby causing the mobius magnet array **630** to rotate in a first direction.

[0041] The shaft **640** is disposed within the outer shell **610**. In some embodiments, only a portion of the shaft **640** is disposed within the outer shell **610** such that a portion of the shaft **640** protrudes from the outer shell **610**. In some embodiments, the shaft **640** is attached to the mobius magnet array **630**. In some embodiments, the shaft **640** is attached to the mobius magnet array **630** such that a portion of the shaft **640** is disposed through the central orifice of the mobius strip of the mobius magnet array **630**. In some embodiments, the shaft **640** is connected to the mobius magnet array **630** such that when the mobius magnet array **630** is rotated by the magnetic fields generated by the first and second external magnet arrays **620a** and **620b** the shaft **640** is also rotated in the same direction as the mobius magnet array **630**.

[0042] In some embodiments, the shaft **640** is further connected to an alternator or other power generation device. The alternator is configured to convert the rotational movement of the shaft **640** into electrical power.

[0043] In some embodiments, the system **600** may further include coils such that the rotational movement of the mobius magnet array generates an electrical current in the coils. This configuration may beneficially combine the features of a prime mover with an alternator.

[0044] FIG. **6B** shows a schematic view of a device **600** for generating power connected to a battery storage system **660**. As seen in FIG. **6B**, the device **600** may be electrically connected to a battery storage system **660**. In some embodiments, the battery storage system **660** is electrically connected to at least one of the external magnet arrays **620a** and **620b**. In some embodiments, the battery storage system **660** is electrically connected to the mobius magnet array **630**. In some embodiments, the battery storage system **660** is electrically connected to an internal magnet array **650** as described below in connection with FIG. **7**. In some embodiments, the battery storage system **660** provides electrical power to the external magnet arrays **620a** and **620b**, the mobius magnet array **630**, and/or the internal magnet array **650**. In some embodiments, the power provided by the battery storage system **660** to the external magnet arrays **620a** and **620b**, the mobius magnet array **630** and/or an internal magnet array **650** augments or initiates the magnetic fields generated by the external magnet arrays **620a** and **620b**, the mobius magnet array **630**, and/or the internal magnet array **650** respectively. This, in turn, causes the mobius magnet array **630** to begin to rotate or to rotate at an increased rate, thereby causing the shaft **640** to begin to rotate or to rotate at an increased rate. This, in turn, leads to initiation of power generation by the device **600** or to an increase in power generation by the device **600**.

[0045] In some embodiments, the battery storage system **660** is further configured to receive electrical power from the device **600**. In some embodiments, a generator or alternator converts the rotational energy of the shaft **640** into electrical power which is then directed to the battery storage system **660**. In some embodiments, only a portion of the power generated by the device **600** and/or the generator is directed to the battery storage system **660**. The power received from the device **600** or from the generator may serve to charge the batteries of the battery storage system **660**.

[0046] Including a battery storage system **660** beneficially allows for “back and forth” transfer of electrical power between the battery storage system **660** and the device **600**. This beneficially allows for the rotational movement of the mobius magnet array **630** and the shaft **640** of the device **600** to be consistently maintained and/or controlled, enables unique form factors, and improves existing electrical generation systems but with greater savings.

[0047] FIG. **7** shows another view of the device **600**. As seen in FIG. **7**, the device **600** may include a plurality of external magnet arrays **620** arranged around the mobius magnet array **630**. As seen in FIG. **7**, the external magnet arrays **620** may be orientated around the mobius magnet array **630** such that the poles of the external magnet arrays **620** located opposite to each other have opposite polarities. In some embodiments, the external magnet arrays **620** may be orientated around the mobius magnet array **630** such that the south magnetic pole of a first magnet array is closest in proximity to the south magnetic pole of an adjacent external magnet array.

[0048] As seen in FIG. **7**, in some embodiments, the device **600** further includes at least one internal magnet array **650**. In some embodiments, the internal magnet array **650** is disposed within a central orifice of the mobius magnet array **630**. In some embodiments, the internal magnet array **650** is stationary does not move with the mobius magnet array **630**. In some embodiments, the internal magnet array **650** is attached to mobius magnet array **630** and rotates with the mobius magnet array **630**. In some embodiments, the internal magnet array **650** creates a magnetic field with the outer shell **610** and causes the mobius magnet array **630** to rotate. In some embodiments, the internal magnet array **650** is disposed with a stationary set of conductors. In some embodiments, the rotational movement of the internal magnet array **650** within the conductors creates electrical current.

[0049] FIG. 8A shows a schematic view of a system **800** for generating power.

[0050] The system may include a plurality of devices. In some embodiments, the system **800** includes a first device **810** and a second device **850**.

[0051] In some embodiments, the first device **810** is similar or identical to the device **600** for generating power as described above. The first device **810** may include an outer shell, at least one external magnet array **820**, mobius magnet array **830** and a shaft **840** similar or identical to the device **600**. In some embodiments, the plurality of magnets of the mobius magnet array **830** are permanent magnets.

[0052] In some embodiments, the first device **810** is electrically connected to the second device **850**. In some embodiments, the shaft **840** of the first device **810** is connected to an alternator which is electrically connected to the second device **850**. The alternator converts the rotational energy of the shaft **840** into electrical power. In some embodiments, the electrical power generated by the first device **810** is directed to the second device **850**.

[0053] In some embodiments, the second device **850** is similar or identical to the device **600** for generating power as described above. The second device **850** may include an outer shell, at least one external magnet array **870**, a mobius magnet array **880** and a shaft **890** similar or identical to the device **600**. In some embodiments, the first device **810** is similar to the second device **850**. In some embodiments, the second device **850** is larger than the first device **810**. In some embodiments, the plurality of magnets of the mobius magnet array **880** of the second device **850** are electromagnets. In some embodiments, the electromagnets of the mobius magnet array **870** are powered by the electrical power generated by the first device **810**.

[0054] In some embodiments, the shaft **890** of the second device **810** is connected to an alternator. The alternator is configured to convert the rotational energy of the shaft **890** into electrical power.

[0055] In some embodiments, the power generated by the second device **850** may be greater than the electrical power generated by the first device **810**.

[0056] FIG. 8B shows a schematic view of a system **800** for generating power. As seen in FIG. 8B, the system **800** further includes a battery storage system **805**. In many respects, the battery storage system **805** may be similar or identical to the battery storage system **600** described above. The battery storage system **805** may be electrically connected to the first device **810** or second device **850**. In some embodiments, the battery storage system is electrically connected to both the first device **810** and the second device **850**.

[0057] In some embodiments, the battery storage system **805** is electrically connected to the at least one external magnet array **820** of the first device **810**. In some embodiments, the battery storage system **805** provides electrical power to the external magnet array **820** of the first device **810**. In some embodiments, the power provided by the battery storage system **805** to the external magnet array **820** initiates or increases the rotational movement of the mobius magnet array **830** and the shaft **840**.

[0058] In some embodiments, the battery storage system **805** is electrically connected to the at least one external magnet array **870** of the second device **850**. In some embodiments, the battery storage system **805** is electrically connected to the mobius magnet array **880** of the second device **805**. In some embodiments, the battery storage system **805** provides electrical power to the external magnet array **870** and/or the mobius magnet array **880** of the second device **850**. In some embodiments, the power provided by the battery storage system **805** to the external magnet array **870** and/or the mobius magnet array **880** of the second device **850** supplements the power provided by the first device **810** to the second device **850**, thereby augmenting the magnetic field generated by the external magnet array **870** and/or the mobius magnet array **880**. This, in turn, causes increased rotational movement of the mobius magnet array **880** and the shaft **890** thereby increasing the electrical power generation of the second device **850**. In some embodiments, the power provided by the battery storage system **805** to the external magnet array **870** and/or the mobius magnet array **880** initiates the rotational movement of the mobius magnet array **880** and the shaft **890**.

[0059] In some embodiments, the battery storage system **805** is further configured to received electrical power from the second device **850**. In some embodiments, a generator or alternator converts the rotational energy of the shaft **890** into electrical power which is then directed to the battery storage system **805**. In some embodiments, only a portion of the power generated by the second device **850** and/or the generator is directed to the battery storage system **805**. The power received from the second device **850** or from the generator may serve to charge the batteries of the battery storage system **805**.

[0060] Including a battery storage system **805** beneficially allows for “back and forth” transfer of electrical power between the battery storage system **805** and the system **800**. This beneficially allows for the rotational movement within the first device **810** and second device **850** to be consistently maintained and/or controlled, enables unique form factors, and improves existing electrical generation systems but with greater savings.

[0061] FIG. **9A** shows a schematic view of a system **900** for powering a home **910**. The home **910** receives power from a variety of power sources. The power sources may include an electrical grid **920**, a battery backup **922**, a standby generator **926**, and solar panels **924**. Each of the power sources may provide power to the home **910**. The power received from the various power sources are used to power various parts of the home **910** including lighting, kitchen appliances, laundry appliances, electronics, HVAC, electronics, and user devices. The home **910** includes a control box **912**. The control box **912** is configured to control or manage the amount of electrical power received from the power sources. The control box **912** is further configured to control which power sources the home **910** receives power from. The control box may direct power from the grid **920** to the battery backup **922** in order to charge the batteries. The control box **912** may also direct any excess power generated by the power sources to the electrical grid **920** in exchange for payment.

[0062] FIG. **9B** shows a schematic view of a system **900** including a power generation system **930**. In some embodiments, the power generation system **930** is similar or identical to the power generation device **600** as described herein or the power generation system **800** as described herein. As seen in FIG. **9B**, a home **910** may be configured to receive electrical power generated by the power generation system **930**. The home **910** may include a control box **912** configured to direct power received from the power generation system **930** to various parts of the home **910**. In some embodiments, the control box **912** may include a variety of hardware, including graphics cards (GPUs), microprocessors (CPUs), storage hard-drives (HDs), and RF (WiFi) devices. In some embodiments, the control box **912** is configured to manage the uptime of the hardware and/or synchronize with blockchain networks, such as the Theta Network. In some embodiments, the hardware of the control box **912** is modularized. This configuration is beneficial for allowing the hardware to be upgraded and/or updated periodically. In some embodiments, the hardware may be not a part of the control box **912**.

[0063] In some embodiments, the control box **912** is configured to direct any excess power generated by the power generation system **930** to the hardware within the control box **912**. In some embodiments, the hardware uses the excess energy provided by the power generation system **930** to supply computational power to a blockchain network. The computational power may be used to mine and/or validate reward based blockchains, such as Bitcoin and Theta Network. For example, the Theta Network rewards participants a reward cryptocurrency, Theta Fuel (TFUEL), for allocating unused computational capacity from GPUs, CPUs, Storage HD, as well as unused internet bandwidth. The Theta Network offers this collective excess hardware capacity as useable cloud and internet bandwidth services to their customers. This power generation system as described herein beneficially allows a user to receive rewards such as payment or cryptocurrency in exchange for the excess power generated by the power generation system **930**. In some embodiments, the control box **912** is further configured to direct excess power generated by the power generation system **930** to an electrical power grid in exchange for payment.

[0064] In some embodiments, payment received in the form of cryptocurrency can be stored in a

wallet locally within the hardware of the control box **912** or can be stored in a wallet managed by a 3rd party provider. Access and visibility to the wallet can be managed through a consumer's mobile device. The cryptocurrency earnings in the wallet can be automatically exchanged for local currency like dollars or spent as cryptocurrency.

[0065] In some embodiments, the system **900** of FIG. **9B** may further include additional power sources including solar panels **924** and/or a battery storage **922** as seen in FIG. **9C**. As seen in FIGS. **9C** and **9D**, the control box **912** may be further configured to use excess power generated by the solar panels **924**, the power generation system **930** and the battery storage **922** to provide computational power to blockchain networks in exchange for payment or cryptocurrency or to provide electrical power back to an electrical grid in exchange for payment.

[0066] As seen in FIGS. **9A**, **9C**, and **9D**, the system **900** may include multiple power sources. Systems with multiple power sources (for example a battery storage and a mobius magnet array) beneficially increase the “uptime” of hardware connectivity. Consistency or reliability of “up time” for hardware connectivity is important when contributing computational resources to networks like Theta. A home or building with multiple backup power options guarantees high “up time” as power can be supplied to the hardware from a second power source if a first power source fails. The systems described herein beneficially allow a home or building with resilient power backup redundancy to become a competitive and reliable node in a blockchain network.

[0067] FIG. **10** shows a variety of forms that the power generation system **930** as described herein can be implemented into. As seen in FIG. **10**, the power generation system **930** as described herein can be implanted into a home **910** or residential dwelling, into a vehicle **940**, such as, but not limited to, an automobile, watercraft, or aircraft, into a user device **950**, such as, but not limited to, mobile devices, personal computers, wearable devices, or into other buildings **960**. The power generation system **930** as described herein can be scaled in size for commercial industrial use or miniaturized for cell phone and/or vehicle power generation.

[0068] The methods disclosed herein comprise one or more steps or actions for achieving the described method. The method steps and/or actions may be interchanged with one another without departing from the scope of the claims. In other words, unless a specific order of steps or actions is required for proper operation of the method that is being described, the order and/or use of specific steps and/or actions may be modified without departing from the scope of the claims.

[0069] It should be noted that the terms “couple,” “coupling,” “coupled” or other variations of the word couple as used herein may indicate either an indirect connection or a direct connection. For example, if a first component is “coupled” to a second component, the first component may be either indirectly connected to the second component or directly connected to the second component. As used herein, the term “plurality” denotes two or more. For example, a plurality of components indicates two or more components.

[0070] Headings are included herein for reference and to aid in locating various sections. These headings are not intended to limit the scope of the concepts described with respect thereto. Such concepts may have applicability throughout the entire specification.

[0071] The previous description of the disclosed implementations is provided to enable any person skilled in the art to make or use the present disclosure. Various modifications to these implementations will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other implementations without departing from the spirit or scope of the disclosure. Thus, the present disclosure is not intended to be limited to the implementations shown herein but is to be accorded the widest scope consistent with the principles and novel features disclosed herein.

Claims

1. A device for generating power, the device comprising: a mobius magnet array comprising: a surface in the form of a mobius strip; and a plurality of magnets disposed on the surface; an external magnet array configured to generate a magnetic field within the device, wherein the magnetic field exerts a rotational force on the mobius magnet array; a shaft connected to the mobius magnet array, the shaft configured to rotate when the magnetic field exerts a force on the mobius magnet array; and an alternator connected to the shaft, the alternator configured to convert the rotational motion of the shaft into electrical power.
 2. A method for generating power, the method comprising: providing a mobius magnet array comprising: a surface in the form of a mobius strip; and a plurality of magnets disposed on the surface; exposing the mobius magnet array to an external magnetic field; rotating the mobius magnet array in response to the external magnetic field; rotating a shaft connected to the mobius magnet array; and converting the rotational motion of the shaft into electrical power.
 3. The method of claim 2, further comprising: directing at least a portion of the electrical power to hardware; providing computational power, via the hardware, to a blockchain network; and validating a blockchain within the blockchain network.
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